

TED (15)6036
Revision 2015

Reg: No.....
 Signature.....

**DIPLOMA EXAMINATION IN ELECTRICAL
 AND ELECTRONIC ENGINEERING- MODEL QUESTION PAPER
 INDUSTRIAL AUTOMATION**

Time: 3 hours.
 (Maximum marks: 100)

PART—A

I. (Answer the following questions in one or two sentences. Each question carries 2 marks.)
 5x2=10

1. State the needs for automation
2. Define closed loop controller
3. Specify a typical input switch
4. What is a servo system
5. Define DCS

PART-B

II (Answer any five of the following. Each question carries 6 marks) 5x6=30

1. Write the importance of automation in terms of reliability and precision
2. Explain the control system
3. Choose a suitable 3 pole contactor for supplying a 5 kW induction load
4. Explain the working of a pull type solenoid
5. Draw the symbols of the following
 - a) Circuit interrupter
 - b) Foot switch
 - c) Liquid level switch
 - d) Pilot lamp
 - e) 3φ squirrel cage motor
 - f) Annunciator
6. Explain the synchros as an error detector
7. Recognize various architectures of SCADA

PART—C

(Answer one full question from each unit. Each question carries 15 marks) 4x15=60

MODULE—I

- III .a) Draw and explain the block diagram of open loop control system . 7
- b) Demonstrate how controlled variable is kept at set point in a closed loop control system 8

OR

- IV .a) Compare advantages of closed loop control system with open loop system 7
- b) Develop the mimic diagram of a temperature controller 8

MODULE—II

- V .a) Propose a suitable circuit diagram for Hoist Control 7
- b) Explain the working principle of bimetallic type temperature sensor 8

OR

- VI.a) Draw a relay (110v) control circuit for controlling the 415 V AC motor which drives a Simple conveyor system having following controls and facilities 1. ON-OFF control Separate pushbutton 2. Overload control 3. Pilot lamp indication on tripping the motor 7
- a) Discuss the working of a control valve 8

MODULE—III

- VII a) Compare the stepper motor & servomotor mechanism 7
- b) What is meant by control transformer in servomechanism 8

OR

- VIII a) Explain the working of synchros 7
- b) Compare the microprocessor and micro controller 8

MODULE-IV

- IX a) Explain pneumatic PID controller 7
- b) What is RTU? Explain its functions. 8

OR

- X a) Explain DCS and its merits 7
- b) What are the applications of SCADA 8
